Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech | *Time: 03 Hours*
Code: 100304 | *Subject: Data Structure & Algorithms* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin
(ii) There are NINE questions in this paper
(iii) Attempt FIVE questions in all
(iv) Question No. 1 is compulsory

Q.1 Choose the correct option / answer from the following (Any seven question only):- [2 x 7 = 14]

(a) Which of the following is not a linear data structure?
- (i) Stack
- (ii) Array
- (iii) Queue
- (iv) Tree

(b) Which asymptotic notation gives the best case complexity?
- (i) Big Theta
- (ii) Big Omega
- (iii) Big O
- (iv) Little o

(c) In stack, which operation is not allowed directly?
- (i) Push
- (ii) Peek
- (iii) Pop
- (iv) Deletion from bottom

(d) Which data structure is used in recursive function calls?
- (i) Queue
- (ii) Linked List
- (iii) Stack
- (iv) Array

(e) Which queue allows insertion and deletion at both ends?
- (i) Circular queue
- (ii) Deque
- (iii) Simple queue
- (iv) Priority queue

(f) Which of the following expression is in postfix form?
- (i) A + B
- (ii) A B C * +
- (iii) A + B * C
- (iv) (A + B) * C

(g) Which of the following has both forward and backward traversal?
- (i) Singly Linked List
- (ii) Circular Queue
- (iii) Doubly Linked List
- (iv) Simple Queue

(h) Which of the following trees is height-balanced?
- (i) Binary Tree
- (ii) AVL Tree
- (iii) B -Tree
- (iv) Threaded Tree

(i) What is the time complexity for insertion at the beginning of a singly linked list?
- (i) O(1)
- (ii) O(n)
- (iii) O(log n)
- (iv) O(n log n)

(j) Which of the following uses divide and conquer strategy?
- (i) Insertion Sort
- (ii) Merge Sort
- (iii) Bubble Sort
- (iv) Linear Search

Q.2 (a) Assume the declaration of multi-dimensional arrays A and B to be, A (-2:2, 2:22) and B (1:8, -5:5, -10:5) [7]

(i) Find the length of each dimension and number of elements in A and B.

(ii) Find the address of element B (2, 2, 3), assuming Base address of B =400 and there are W=4 words per memory location.

(b) What is Sparse matrix? Explain how a Sparse matrix can be implemented by using the linked list? [7]

Q.3 (a) Write an algorithm for Insertion Sort. Use Insertion sort algorithm, sort the following elements: [7]

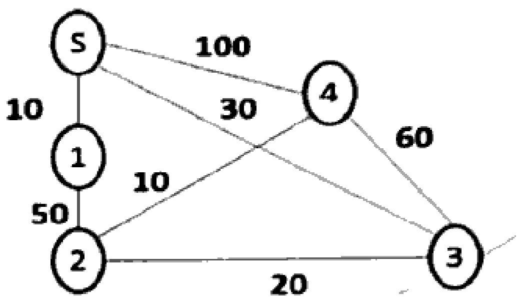
2, 8, 7, 1, 3, 5, 6, 4.

(b) What is circular linked list? Write an algorithm to delete a node from begin in single linked list. [7]

Q.4 (a) Consider the following infix expression and convert it into postfix using stack [7]

A + (B * C – (D/E-F) * G) * H

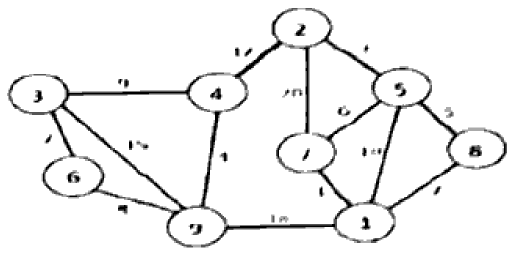
(b) Find the shortest path from S to all remaining vertices of graph using Dijikstra Algorithm [7]



Q.5 (a) Define Hashing. Explain the mid-square and digit folding method hash function with the help of an example. [7]

(b) Use Heap sort algorithm to sort the following sequence {8, 5, 45, 24, 36, 11, 43, and 21}. [7]

Q.6 (a) Discuss spanning tree. Write down the Kruskal algorithm to obtain a minimum cost spanning tree. Use Kruskal algorithm to find the minimum cost spanning tree in the following graph: [7]



(b) What do you mean by priority queue? Explain the types to maintain the priority queue in memory? [7]

Q.7 (a) Explain B-tree. Write down the properties of it. Consider following sequence of inputs: [7]

10,20,35,40,50,60,75,80,95

Assume that the order of the B-tree is 3.

(b) Draw a binary tree with following traversals: [7]

Preorder: A B C D E F G H I J K L

Postorder: C F E G D B K J L I H A

Q.8 (a) Do the following operations for constructing a BST [7]
(i) 45, 37, 98, 76, 13, 39,105, 80, 5 insert element as per their occurrence.
(ii) Delete 39 and 45 respectively
Now Traverse the final BST in Inorder, Preorder and Postorder.

(b) Describe an AVL tree. Construct an AVL tree by inserting the following elements in the order of their occurrence {60, 2, 15, 20, 12, 115, 90 and 88}. [7]

Q.9 (a) Write merge sort algorithm and its analysis. Use merge sort algorithm to sort 9, 11, 10, 1, 60, 10, 6, 25, 40, and 30. Is it a stable sorting algorithm? Justify. [7]

(b) Write short notes on ***any one*** the following: [7]
(i) Double-ended Queue (Deque)
(ii) Threaded Binary Tree
(iii) Time Complexity and Big-O Notation
(iv) DFS, BFS